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Roll No.....

TX-503 (GS)

B.E. V Semester Examination, June 2020

Grading System (GS)

Measurement and Control

Time : Three Hours

Maximum Marks : 70

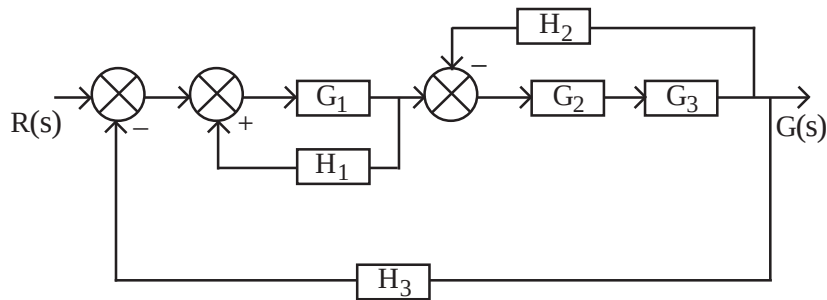
Note: i) Attempt any five questions.

ii) All questions carry equal marks.

1. A 0-150 V voltmeter has a guaranteed accuracy of 1% of full scale reading. The voltage measured by this instrument is 75 V. Calculate the limiting error in percentage. Also comment upon the result.
2. Describe the method for measurement of temperature with the use of
 - i) RTD
 - ii) Bimetallic thermometer
3. Discuss the following:
 - i) Error Propagation
 - ii) Bias Error
 - iii) Precision Error
 - iv) Data Acquisition
4. Write short note on:
 - i) Replication and Repetition
 - ii) Sequential and Random Test
 - iii) Noise and Interference
5. Define Noise. Explain various sources of noise with example. Describe the techniques used for reducing the magnitude of noise.
6. a) Explain calibration with their type.
b) Discuss DVM. Compare Analog and Digital Instruments.

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7. a) Discuss Seven Segment Display.
b) Solve Using Block Diagram Reduction Method.



8. Define the following terms:(Any two)

- i) Orifice plate
- ii) Accuracy
- iii) Odds
- iv) Replication and repetition
